Disentangling Good-Enough and Rational Inference in Sentence Illusions

Jiaxuan Li & Richard Futrell (University of California Irvine)

Introduction: Language comprehension can deviate from the literal meaning of the input. Given "*the mother gave the candle the daughter*", comprehenders might arrive at a non-literal interpretation where the daughter received the candle. This non-literal interpretation process has been generally associated with an interaction between prior predictions based on world knowledge and bottom-up perceptual input. When there is a great overlap between prior prediction and present input, the priors can override presented input so that comprehenders interpret the sentence non-literally. However, the nature of the interactive process has been under debate. According to **good-enough** (GE) account [1-2], strong contextual expectations can distort the perception of linguistic input, and that comprehenders might process the sentence superficially and perceive it as a good-enough alternative. Alternatively, the non-literal interpretation could arise because of an effortful **rational inference** (RI) process [3-5]. Priors influence interpretation after comprehenders have established an accurate representation of the presented input. Comprehenders rationally infer the speaker's intended meaning given the presented input, and may error-correct the perceived form into a more plausible alternative.

Hypothesis: We tease apart GE and RI by examining the proportion of non-veridical interpretations with **constrained cognitive resources**. GE predicts non-veridical interpretation is more likely when the cognitive resources are limited. RI predicts comprehenders are more likely to arrive at non-veridical interpretation with enough cognitive resources. Moreover, it is possible that both strategies are used (hybrid account [6]), and therefore non-literal interpretations are more likely when cognitive resources are either very limited or unrestricted.

Experiment: We adapted 60 sets of English Double Object (DO) and Prepositional Object (PO) sentences and 60 fillers from [7] . We used a 2×3 within-subject design manipulating **sentence type** (illusion vs. control; see List 1) and **presentation speed** (fast, medium, slow; List 2). Each subject viewed either the illusion or control version of each sentence set (60 sets + 60 fillers). After the presentation of a fixation cross for 1000ms, sentences were presented one word at a time. The stimuli are distributed into three blocks, and fully randomized within each block. Each block is presented either in a fast / normal or slow speed. Presentation duration scaled with word length. After each sentence, participants answered a yes/no question probing whether they adopted the literal or non-literal interpretation (e.g. *"Did the daughter receive something?"*). Native English speakers (ages 18–45) were recruited through Prolific (N=62). As shown in Fig. 1, for illusion sentences, GE predicts highest rate of non-literal interpretation in fast presentation rate; RI predicts highest rate of non-literal interpretation in slow condition; whereas hybrid account makes non-monotonic or subject-specific predictions.

Analysis and Results: We recruited 62 subjects from Prolific. Subjects with average accuracy below 80% on comprehension questions of filler sentences in normal and slow presentation blocks are removed, yielding 49 subjects. Trials with RT outside 3 std of mean RT averaged within each subject and within each block were further removed. Responses were modeled using a linear mixed-effects model (Eq. 1). As shown in Fig. 2, there is a robust effect of sentence type ($beta = -2.81$, $p < .001$), suggesting that comprehenders are more accurate in control than in illusion condition. There is a significant effect of presentation speed ($beta = -0.64$, $p < 0.01$), indicating that comprehenders are more likely to adopt literal interpretation in slow presentation rate. This supports the good-enough processing account. However, visual inspection across subjects suggests great between-subject variation. There is no significant interaction between sentence type and presentation speed ($beta = 0.005$, $p = 0.997$), indicating that presentation speed did not modulate the magnitude of this illusion.

Conclusion Our findings show that non-literal interpretations are more frequent at fast presentation speeds, supporting the good-enough account in which reduced cognitive efficiency increases susceptibility to misperception, though substantial individual variability suggests that comprehenders may differ in their reliance on heuristic versus inference-based strategies.

List 1. Sample Stimuli

- **Illusion:** *the mother gave the candle the daughter*
- **Control:** *the mother gave the daughter the candle*

List 2. Word presentation rate
Word duration = base rate × 100 ms + 5 ms × word length.

- **Fast:** base = 1 (105–175 ms/word, ISI 0 ms)
- **Medium:** base = 4 (405–475 ms/word, ISI 200 ms)
- **Slow:** base = 8 (805–875 ms/word, ISI 400 ms)

Equation 1. Correctness of comprehension question ~ speed * sentence type + (1+ speed * sentence type | subject) + (1+ speed * sentence type| item)

Fig. 1. Average proportion of literal interpretations across different presentation speeds.

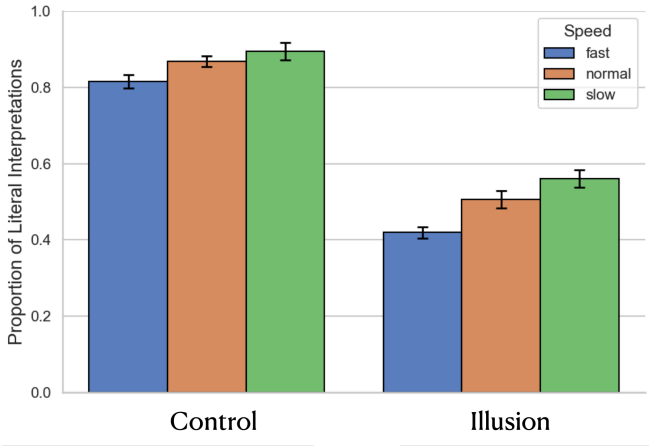
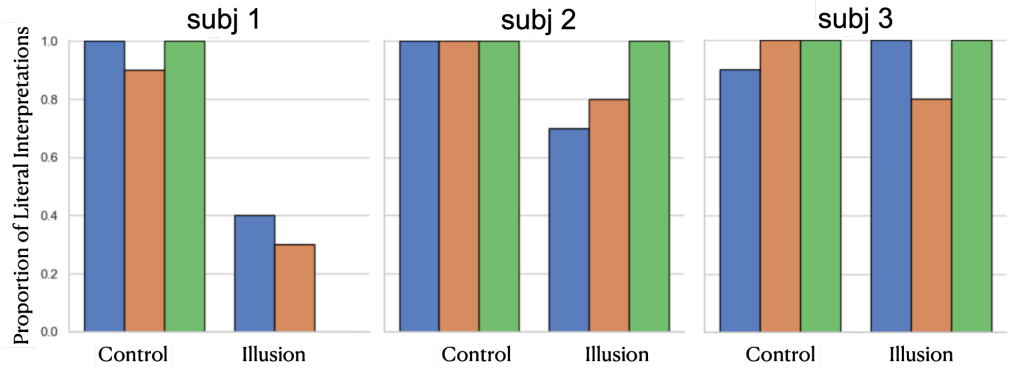


Fig. 2. Proportion of literal interpretations across three selected subjects



References [1] Ferreira & Patson (2007). *Language and linguistics compass*. [2] Ferreira, Bailey & Ferraro (2002). *Current directions in psychological science*. [3] Gibson, Bergen, & Piantadosi (2013). *PNAS*. [4] Levy et al. (2009). *PNAS.* [5] Ryskin et al. (2018). Comprehenders model the nature of noise in the environment. *Cognition*. [6] Paape, D. (2024). *Language, Cognition and Neuroscience*. [7] Cai., Zhao & Pickering (2022). *Cognition*.